Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

SEMI E87.1-0225
SPECIFICATION FOR SECS-II PROTOCOL FOR CARRIER MANAGEMENT (CMS)

This Standard was technically approved by the Information & Control Global Technical Committee. This edition was approved for publication by the global Audits and Reviews Subcommittee on December 10, 2024. Available at www.semiviews.org and www.semi.org in February 2025; originally published February 2000; previously published May 2024.

NOTICE: The section symbol (§) is used to reference a numbered section, paragraph, or header and all subordinate paragraphs, headers, and embedded material (i.e., EXCEPTIONs, lists) therein. The paragraph symbol (¶) is used to reference a particular numbered paragraph and embedded material therein. When duplicated (i.e., §§ and ¶¶) the symbols are used to reference more than one section or paragraph, respectively.

NOTICE: Paragraphs entitled 'NOTE:' are not an official part of this Standard or Safety Guideline and are not intended to modify or supersede the official Standard or Safety Guideline. These have been supplied by the global technical committee to enhance the usage of the Standard or Safety Guideline.

1 Purpose

1.1 This Specification maps the services and data of SEMI E87 to SECS-II streams and functions, and data definitions.

2 Scope

2.1 This Specification covers equipment supporting automated access to load ports from the host point-of-view.

2.2 This Specification applies to all implementations of SEMI E87 that use the SECS-II message protocol (SEMI E5). Compliance to this Standard requires compliance to both SEMI E87 and SEMI E5.

NOTICE: SEMI Standards and Safety Guidelines do not purport to address all safety issues associated with their use. It is the responsibility of the users of the Documents to establish appropriate safety and health practices, and determine the applicability of regulatory or other limitations prior to use.

3 Limitations

3.1 This Specification applies to semiconductor equipment with SEMI E15.1 compliant load ports. It may also be applied to other manufacturing equipment that supports automated carrier transfer and or contains an internal buffer.

4 Referenced Standards and Documents

4.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E15.1 — Specification for 300 mm Tool Load Port

SEMI E39.1 — Specification for SECS-II Protocol for Object Services Standard (OSS)

SEMI E87 — Specification for Carrier Management (CMS)

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

5 Services Mapping

5.1 This section shows the specific SECS-II streams and functions that shall be used for SECS-II implementation of the services defined in SEMI E87, as well as the parameter mapping for data attached to services.

5.2 *Services Message Mapping*

5.2.1 Table 1 defines the relationships between SEMI E87 services and SECS-II messages.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Table 1 Services Message Mapping Table

Service Name	*Stream,Function*	*SECS-II Message Name*
Bind	S3,F17/18	Carrier Action Request/Acknowledge
CancelBind	S3,F17/18	Carrier Action Request/Acknowledge
CancelAllCarrierOut	S3,F19/20	Cancel All Carrier Out Request/Acknowledge
CancelCarrier	S3,F17/18	Carrier Action Request/Acknowledge
CancelCarrierAtPort	S3,F17/18	Carrier Action Request/Acknowledge
CancelCarrierNotification	S3,F17/18	Carrier Action Request/Acknowledge
CancelCarrierOut	S3,F17/18	Carrier Action Request/Acknowledge
CancelReservationAtPort	S3,F25/26	Port Action Request/Acknowledge
CarrierIn	S3,F17/18	Carrier Action Request/Acknowledge
CarrierNotification	S3,F17/18	Carrier Action Request/Acknowledge
CarrierOut	S3,F17/18	Carrier Action Request/Acknowledge
CarrierReCreate	S3,F17/18	Carrier Action Request/Acknowledge
CarrierRelease	S3,F17/18	Carrier Action Request/Acknowledge
CarrierTagReadData	S3,F29/30	Carrier Tag Read Data Request/Acknowledge
CarrierTagWriteData	S3,F31/32	Carrier Tag Write Data Request/Acknowledge
ChangeAccess	S3,F27/28	ChangeAccess
ChangeServiceStatus	S3,F25/26	Port Action Request/Acknowledge
ProceedWithCarrier	S3,F17/18	Carrier Action Request/Acknowledge
ReserveAtPort	S3,F25/26	Port Action Request/Acknowledge

5.3 *Services Parameter Mapping*

5.3.1 Table 2 maps the SEMI E87 service parameters to SECS-II Data Items.

NOTE 1: Use of parameters not specified for a given message in SEMI E87 is prohibited. SECS-II data items not used for a given message shall be sent as zero-length items.

Table 2 Service Parameters to SECS-II Data Items Mapping

Parameter Name	*Range*	*SECS-II Data Item*
AccessMode	Enumerated: MANUAL, AUTO	ACCESSMODE
AttributeData	Any	CATTRDATA
AttributeID	Text format restrictions per SEMI E39.1, § 6.	CATTRID
CarrierID	1 to 80 characters	CARRIERID
CMAcknowledge	Enumerated	CAACK
CMStatus	Structure	L,2 1. <CAACK> 2. Status
Data	ASCII (Format Code 20)	DATA
DataLength	Integer (Format Code 30–34 or 50–54)	DATALENGTH
DataSeg	ASCII (Format Code 20)	DATASEG
ErrorCode	Enumerated	ERRCODE
ErrorText	1 to 80 characters	ERRTEXT
LocationID	ASCII (Format Code 20)	LOCID

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Parameter Name	*Range*	*SECS-II Data Item*
PropertiesList	Non-identifier properties n = number of attributes	L,n 1. L,2 1. <CATTRID$_1$> 2. <CATTRDATA$_1$> … n. L,2 1. <CATTRID$_n$> 2. <CATTRDATA$_n$>
PortID	Format Code 51 (1–255)	PTN
ServiceStatus	Enumerated: IN SERVICE, OUT OF SERVICE	Format Code 51 0 = OUT OF SERVICE 1 = IN SERVICE
Status	n = number of errors	L,n 1. L,2 1. <ERRCODE$_1$> 2. <ERRTEXT$_1$> … n. 1. <ERRCODE$_n$> 2. <ERRTEXT$_n$>

5.4 *SECS-II Data Items Without Corresponding SEMI E87 Parameters*

5.4.1 Table 3 contains the SECS-II data items that do not correspond to SEMI E87's service parameter.

Table 3 Additional Data Item Requirements Table

Function	*SECS-II Data Item*
Used by S3,F17 to differentiate between Bind, CancelCarrierOut, CancelCarrierAtPort, CancelBind, CarrierIn, ProceedWithCarrier, CancelCarrierNotification CarrierNotification, CarrierReCreate, CancelCarrier, CarrierOut and CarrierRelease services.	CARRIERACTION
Used to satisfy SECS-II conventions for linking a multi-block inquiry with a subsequent multi-block message. Neither required nor specified by CMS.	DATAID
Used to inform receiver of total message length size for SECS-II multi-block conventions. May also be used to indicate the length of a section of data being transmitted to or from a carrier tag.	DATALENGTH
Used to satisfy SECS-II multi-block requirements. Neither required nor specified by SEMI E87.	GRANT
Used by S3,F25 to differentiate between port related, CancelReservationAtPort, and ReserveAtPort services.	PORTACTION
Used by S3,F27 to specify desired Port Access Mode.	ACCESSMODE

6 Variable Data Item Mapping

6.1 This section shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E87 variable data items.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Table 4 Variable Data Item Mapping Table

Variable Name	*Class*	*Format*
AccessMode	DVVAL	Format Code 51 Enumerated: 0 = MANUAL 1 = AUTO
AccessMode$_i$	SV	Format Code 51 Enumerated: 0 = MANUAL 1 = AUTO
AdditionalInfo	DVVAL	L,n where n=number of elements for carrier's AdditionalInfo attribute 1. L,2 1. AdditionalInfoName$_1$ 2. AdditionalInfoValue$_1$ … n. L,2 1. AdditionalInfoName$_n$ 2. AdditionalInfoValue$_n$ AdditionalInfoName$_i$ is Format Code 20, and is limited to 1–256 characters. AdditionalInfoValue$_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
AdditionalSubstrateInfoMap	DDVAL	L,n where n = Capacity 1. L,m where m=number of elements for Substrate 1's AdditionalInfo attribute 1. L,2 1. SubstrateAdditionalInfoName$_1$ 2. SubstrateAdditionalInfoValue$_1$ … m. L,2 1. SubstrateAdditionalInfoName$_m$ 2. SubstrateAdditionalInfoValue$_m$ … n. L,p where p=number of elements for Substrate n's AdditionalInfo attribute 1. L,2 1. SubstrateAdditionalInfoName$_1$ 2. SubstrateAdditionalInfoValue$_1$ … m. L,2 1. SubstrateAdditionalInfoName$_p$ 2. SubstrateAdditionalInfoValue$_p$ SubstrateAdditionalInfoName$_i$ is Format Code 20, and is limited to 1–256 characters. SubstrateAdditionalInfoValue$_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
AvailPartitionCapacity	DVVAL	Format Code 51
AvailPartitionCapacity$_i$	SV	Format Code 51
BufferCapacityList	SV	L,n where n = number of partitions 1. <BufferPartitionInfo$_1$> ... n. <BufferPartitionInfo$_n$>

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Variable Name	*Class*	*Format*
BufferPartitionInfo	DVVAL	L,5 1. <PartitionID> 2. <PartitionType> 3. <AvailPartitionCapacity> 4. <PartitionCapacity> 5. <UnallocatedPartitionCapacity>
BufferPartitionInfo$_i$	SV	L,5 1. <PartitionID$_i$> 2. <PartitionType$_i$> 3. <AvailPartitionCapacity$_i$> 4. <PartitionCapacity$_i$> 5. <UnallocatedPartitionCapacity$_i$>
BypassReadID	EC	Format Code 11
CarrierAccessingStatus	DVVAL	Format Code 51 Enumerated as: 0 = NOT ACCESSED 1 = IN ACCESS 2 = CARRIER COMPLETE 3 = CARRIER STOPPED
CarrierID	DVVAL	Format Code 20, 1 to 80 characters Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
CarrierID$_i$	SV	Format Code 20, 1 to 80 characters Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
CarrierIDStatus	DVVAL	Format Code 51 Enumerated as: 0 = ID NOT READ 1 = [ID] WAITING FOR HOST 2 = ID VERIFICATION OK 3 = ID VERIFICATION FAILED
CarrierObjType	DVVAL	Format Code 20, 1 to 40 characters
CarrierLocationMatrix	SV	L,n where n = number of carrier locations 1. L,2 1. <LocationID$_1$> 2. <CarrierID$_1$> … n. L,2 1. <LocationID$_n$> 2. <CarrierID$_n$>
CarrierMovement Interval	SV	Format Code 54
CCPActualTime	DVVAL	Format Code 20 Formatted per TIMESTAMP SECS-II data item.
CCPActualTimeDifference	DVVAL	Format Code 32
CCPFailedReason	DVVAL	Format Code 20, 1 to 80 characters.
CCPImminentLeadTimeInterval	DVVAL	Format Code 52
CCPredictedTime	DVVAL	Format Code 20 Formatted per TIMESTAMP SECS-II data item.
CCPredictedTimei	SV	Format Code 20 When predicted carrier complete time is specified, it is formatted per TIMESTAMP SECS-II data item.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Variable Name	*Class*	*Format*
CCPStatus	DVVAL	Format Code 51 Enumerated as: 0 = PREDICTION DISABLED 1 = NOT PREDICTED 2 = PREDICTING 3 = PREDICTED 4 = CARRIER COMPLETE IMMINENT 5 = PREDICTION COMPLETED 6 = PREDICTION FAILED
CCPTimeoutInterval	DVVAL	Format Code 52
LoadPortReservationState	DVVAL	Format Code 51 Enumerated as: 0 = NOT RESERVED 1 = RESERVED
LoadPortReservationState$_i$	SV	Format Code 51 Enumerated as: 0 = NOT RESERVED 1 = RESERVED
LoadPortReservationStateList	SV	L,n where n = number of load ports 1. <LoadPortReservationState$_1$> … n. <LoadPortReservationState$_N$>
LocationID	DVVAL	Format Code 20, 1 to 80 characters. Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
LocationID$_i$	SV	Format Code 20, 1 to 80 characters. Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
PartitionCapacity	DVVAL	Format Code 51
PartitionCapacity$_i$	SV	Format Code 51
PartitionID	DVVAL	Format Code 20, 1 to 80 characters. Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
PartitionID$_i$	SV	Format Code 20, 1 to 80 characters. Conforms to restrictions of ObjID as specified in SEMI E39.1, § 6.
PartitionType	DVVAL	Format Code 20, 1 to 64 characters.
PartitionType$_i$	SV	Format Code 20, 1 to 64 characters.
PortAssociationState	DVVAL	Format Code 51 Enumerated as: 0 = NOT ASSOCIATED 1 = ASSOCIATED
PortAssociationState$_i$	SV	Format Code 51 Enumerated as: 0 = NOT ASSOCIATED 1 = ASSOCIATED
PortAssociationStateList	SV	L,n where n = number of load ports 1. <PortAssociationState$_1$> … n. <PortAssociationState$_n$>
PortID	DVVAL	Format Code 51
PortID$_i$	SV	Format Code 51

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Variable Name	*Class*	*Format*
PortStateInfo	DVVAL	L,2 1. <PortAssociationState> 2. <PortTransferState>
$PortStateInfo_i$	SV	L,2 1. <$PortAssociationState_i$> 2. <$PortTransferState_i$>
PortStateInfoList	SV	L,n where n = number of load ports 1. <$PortStateInfo_1$> … n. <$PortStateInfo_n$>
PortTransferState	DVVAL	Format Code 51 Enumerated as: 0 = OUT OF SERVICE 1 = TRANSFER BLOCKED 2 = READY TO LOAD 3 = READY TO UNLOAD
$PortTransferState_i$	SV	Format Code 51 Enumerated as: 0 = OUT OF SERVICE 1 = TRANSFER BLOCKED 2 = READY TO LOAD 3 = READY TO UNLOAD
PortTransferStateList	SV	L,n where n = number of load ports 1. <$PortTransferState_1$> … n. <$PortTransferState_n$>
Reason	DVVAL	Format Code 51 Enumerated as: 0 = VERIFICATION NEEDED 1 = VERIFICATION BY EQUIPMENT UNSUCCESSFUL 2 = READ FAIL 3 = IMPROPER SUBSTRATE POSITION 4 = MISSING SLOTMAP ON SCAN BYPASS
SlotMapStatus	DVVAL	Format Code 51 Enumerated as: 0 = SLOT MAP NOT READ 1 = [SLOT] WAITING FOR HOST 2 = SLOT MAP VERIFICATION OK 3 = SLOT MAP VERIFICATION FAILED

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Variable Name	*Class*	*Format*
SlotMap	DVVAL	L,n n = Capacity 1. Format Code 51 enumerated 2. Format Code 51 enumerated 3. Format Code 51 enumerated … n. Format Code 51 enumerated Enumerated as: 0 = UNDEFINED 1 = EMPTY 2 = NOT EMPTY 3 = CORRECTLY OCCUPIED 4 = DOUBLE SLOTTED 5 = CROSS SLOTTED
UnAllocatedPartitionCapacity	DVVAL	Format Code 51
UnAllocatedPartitionCapacity$_i$	SV	Format Code 51

7 SECS-II Attribute Definitions

7.1 *Carrier Object SECS-II Attributes Definitions*

7.1.1 The following are the SECS-II structure definitions for the SEMI E87 Carrier Object.

Table 5 Carrier Object Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
ObjType	Carrier
ObjID	<CARRIERID> (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6.)
AdditionalInfo	L,n where n=number of elements for carrier's AdditionalInfo attribute 1. L,2 1. AdditionalInfoName$_1$ 2. AdditionalInfoValue$_1$ … n. L,2 1. AdditionalInfoName$_n$ 2. AdditionalInfoValues$_n$ AdditionalInfoName$_i$ is Format Code 20, and is limited to 1–256 characters. AdditionalInfoValue$_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Attribute Name	*Attribute Data Form: SECS-II Structure*
AdditionalSubstrateInfoMap	L,n where n = Capacity 1. L,m where m=number of elements for Substrate 1's AdditionalInfo attribute 1. L,2 1. $SubstrateAdditionalInfoName_1$ 2. $SubstrateAdditionalInfoValue_1$ … m. L,2 1. $SubstrateAdditionalInfoName_m$ 2. $SubstrateAdditionalInfoValue_m$ … n. L,p where p=number of elements for Substrate n's AdditionalInfo attribute 1. L,2 1. $SubstrateAdditionalInfoName_1$ 2. $SubstrateAdditionalInfoValue_1$ … m. L,2 1. $SubstrateAdditionalInfoName_p$ 2. $SubstrateAdditionalInfoValue_p$ $SubstrateAdditionalInfoName_i$ is Format Code 20, and is limited to 1–256 characters. $SubstrateAdditionalInfoValue_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
Capacity	Format Code 51, 52, or 54
CarrierAccessingStatus	Format Code 51 CarrierAccessingStatus CarrierAccessingStatus enumerated per Variable CarrierAccessingStatus
CarrierIDStatus	Format Code 51 CarrierIDStatus CarrierIDStatus enumerated per Variable CarrierIDStatus
CCPImminentLeadTimeInterval	Format Code 52 CCPImminentLeadTimeInterval
CCPredictedTime	Format Code 20 CCPredictedTime When predicted carrier complete time is specified, it is formatted per TIMESTAMP SECS-II data item.
CCPStatus	Format Code 51 CCPStatus CCPStatus enumerated per variable CCPStatus
CCPTimeoutInterval	Format Code 52 CCPTimeoutInterval
ContentMap	L,n where n = Capacity 1. L,2 1. Format Code 20 LotID 2. Format Code 20 SubstID … n. L,2 1. Format Code 20 LotID 2. Format Code 20 SubstID SubstID conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6. or L,0 (list of zero)
LocationID	Format Code 20 LocationID LocationID conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6.

Licensed By SEMI to Han Ho, _SEMI_Individuals
Downloaded: 2025-04-25
Single user license only. Copying and Networking prohibited.

Attribute Name	*Attribute Data Form: SECS-II Structure*
SlotMap	L,n where n = Capacity 1. Format Code 51 enumerated … n. Format Code 51 enumerated Enumerated per variable SlotMap
SlotMapStatus	Format Code 51 SlotMapStatus SlotMapStatus enumerated per Variable SlotMapStatus.
SubstrateCount	Format Code 51, 52, or 54 SubstrateCount SubstrateCount Range: 0 to Capacity SubstrateCount Examples: 1, 3, 21, 25
Usage	Format Code 20 Usage Usage is equipment defined, examples: ‘TEST’, ‘DUMMY’, ‘PRODUCT’

NOTICE: SEMI makes no warranties or representations as to the suitability of the Standards and Safety Guidelines set forth herein for any particular application. The determination of the suitability of the Standard or Safety Guideline is solely the responsibility of the user. Users are cautioned to refer to manufacturer’s instructions, product labels, product data sheets, and other relevant literature, respecting any materials or equipment mentioned herein. Standards and Safety Guidelines are subject to change without notice.

By publication of this Standard or Safety Guideline, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any items mentioned in this Standard or Safety Guideline. Users of this Standard or Safety Guideline are expressly advised that determination of any such patent rights or copyrights and the risk of infringement of such rights are entirely their own responsibility.

Copyright by SEMI®, 673 S. Milpitas Blvd., Milpitas, CA 95035. Reproduction of the contents in whole or in part is forbidden without express written consent of SEMI.